SQL Server 2014

Extending the Buffer Pool with Solid-State Drives

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Speaker

- Gianluca Hotz

Consultant and mentor at SolidQ
  Administration, capacity planning, data modeling, development, optimization
  18 years on SQL Server (from 4.21 in 1996)

Interests
  Relational model, DBMS architectures, high availability and disaster recovery

Microsoft MVP SQL Server since 1998

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Buffer Pool Extension

▶ Introduction

- Adding memory easiest way to improve performance
  - Especially for OLTP
- Can be problematic
  - Expensive
  - Maximum number of slots
  - Maximum DIMM size
- Adding new storage less painful
  - Extension to Buffer Pool on SSD

Buffer Pool Extension

▶ How it works

Under memory pressure

Clean pages

can be evicted

can be written to BP extension

Dirty pages

needs to be written to data files

can be written to BP extension

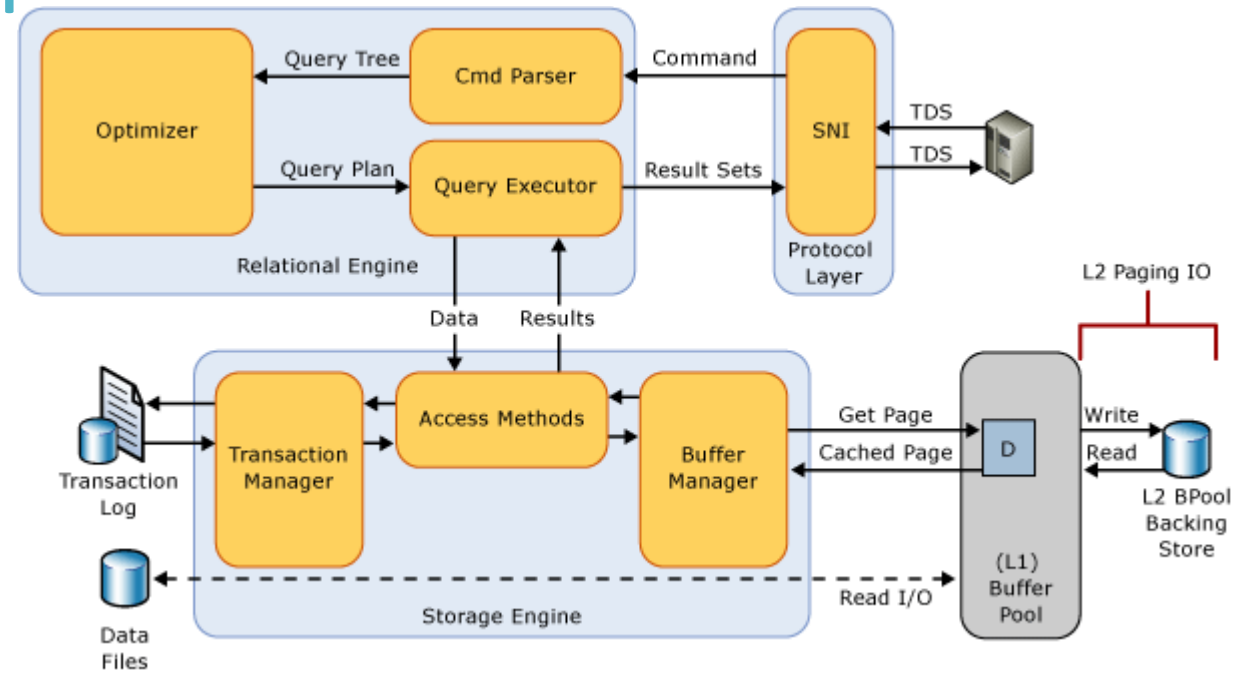

Image source: http://msdn.microsoft.com/en-us/library/dn133176(v=sql.120).aspx

Buffer Pool Extension

- Page usage thresholds

Hot pages in RAM

Warm Pages to BP Extension

Cold Pages evicted

Buffer Pool Extensions

- Configuration

```
ALTER SERVER CONFIGURATION
 SET BUFFER POOL EXTENSION
 { ON ( FILENAME = 'os_file_path_and_name' , SIZE = <size> [ KB | MB | GB ] ) | OFF }
```

  Created during instance startup
  Deleted when instance shut down

Permissions
  ALTER SERVER STATE

Standard Edition (now supports 128GB RAM)

64bit only

Buffer Pool Extension

▶ Configuration Details

- DMV
  - sys.dm_os_buffer_pool_extension_configuration
- Cannot be smaller than “max server memory”
  - Up to 32x “max server memory”
  - Best practice 1:16 (1:4 or 1:8 may be optimal)
- Cannot be changed
  - Needs to be turned off and on again
  - Cannot be smaller than previously configured size
    - *Instance needs to be restarted*

```sql
drop table dbo.STDEntities
drop table dbo.HKTEntities
go

-- Create Standard Table
create table dbo.STDentities
(
    customerid int identity not null,
    attribute1 varchar(100) not null,
    attribute2 varchar(100) not null,
    attribute3 varchar(100) not null,
    constraint pk__STDentities primary key nonclustered(customerid)

)
go

-- Create Hekaton Table
create table dbo.HKTEntities
(
    customerid uniqueidentifier not null default(newid()),
    attribute1 varchar(100) not null,
    attribute2 varchar(100) not null,
    attribute3 varchar(100) not null,
    constraint pk__HKTEntities primary key nonclustered hash (customerid)
        with (bucket_count = 50000)

)
with (memory_optimized = on, durability = schema_only)
go

-- Test simple insert
insert into dbo.STDentities (attribute1, attribute2, attribute3) values ('Some', 'Value', 'Here')
insert into dbo.HKTEntities (attribute1, attribute2, attribute3) values ('Some', 'Value', 'Here')
```

Demo

Buffer Pool Extension

Buffer Pool Extension

▶ Monitoring

Performance Counters

- Extension allocated pages
- Extension free pages
- Extension in use as percentage
- Extension outstanding IO counter
- Extension page evictions/sec
- Extension page reads/sec
- Extension page unreferenced time
- Extension pages writes/sec

Buffer Pool Extension

▶ Troubleshooting

DMV

- sys.dm_os_buffer_descriptors

Extended Events

- buffer_pool_extension_pages_written
- buffer_pool_extension_pages_read
- buffer_pool_extension_pages_evicted
- buffer_pool_eviction_thresholds_recalculated

Buffer Pool Extension

▶ Key Takeaways

Reduce memory pressure with cheaper SSD
- as in “cheaper than RAM” not just cheap!
- use high throughput/endurance disks

No risk of data loss (only clean pages!)

Performance gains for OLTP (mostly read-only)

Limited expectations for
- Data Warehouse workloads
- machines with more than 64GB (max memory for SQL Server)
- write-heavy OLTP workloads

Buffer Pool Extensions

▶ Resources

Video

"Buffer Pool Extension and Resource Governor for IO"
http://channel9.msdn.com/Series/sqlserver-2014-mission-crit-performace/07

ACM SIGMOD Paper

"Turbocharging DBMS Buffer Pool Using SSDs"
http://gsl.azurewebsites.net/Portals/0/Users/Projects/SSD/sigmod2010.pdf

Grazie!

Trovi altri video su:
www.ugiss.org/sql-server-2014-pills